TAFL NORMAND

Strategy and tactics

770 games analysed | 38,307 positions | Classic rules

The advice in this booklet comes from 770 games played between strong computer players (from 60,000 to 1,000,000 positions searched per move) and from targeted analyses. Every figure comes from those games. Computers do not play like humans, but they show what wins when you calculate well.

77%
of wins by the king reaching a corner

22%
by capturing the king

0
by blocking every enemy piece

28
moves per player in a typical game

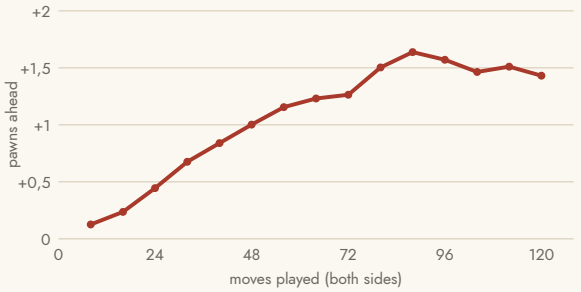
1 How games are won
2 What really matters
3 The opening
4 Moving the pair out
5 Tactical patterns
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1 How games are won

The game stops before the last move. With the threat rule, a king almost never actually enters a corner or gets captured: the opponent must stop the threat. The game ends when no defence is left, an unstoppable win. In 770 games, not a single king was actually captured.

Escapes dominate. 77% of wins come from an unstoppable escape to a corner, 22% from an unstoppable king capture, 1.6% from both at once. Blocking every enemy piece never happens. Only 15 games out of 770 are drawn.

The escape strikes like lightning. In 67% of escape wins, the winning king touches the edge only on the very last move: it jumps from the inside onto a line nobody can close any more.



1 Average lead in pawns of the future winner, by number of moves played (both sides counted).

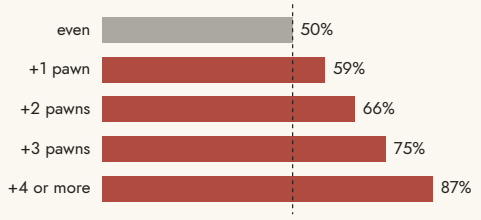
Captures happen in the middle. 85% of captured kings are caught away from the edges, where they can be taken in both directions.

The advantage builds up slowly. The future winner is barely ahead at first: half a pawn around move 24 (counting both players' moves), one pawn around move 48, one and a half later (figure 1).

2 What really matters

To find out what wins, I related the final result of each game to the state of the board in 38,307 positions. The table gives the value of each asset, converted into pawns.

Asset	Value in pawns
One extra pawn	1
King standing on an edge	0.63
Each corner the king can reach in two moves	0.61
An immediate winning threat	0.27
Having the move	0.22
Each square the king can move to	0.16
Guard pawns near the corners	0.06



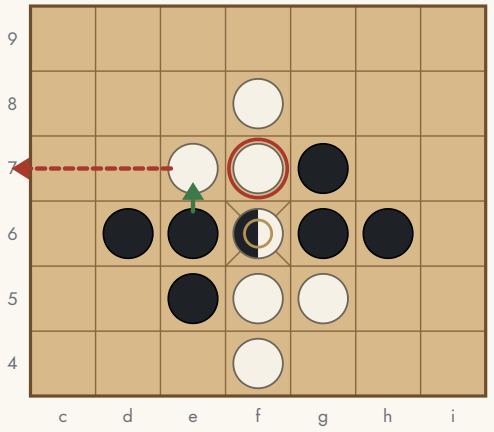
Winning rate by pawn lead (all positions, both sides).

- **Material first.** One extra pawn raises your winning chances from 50% to 59%, two pawns to 66%, four or more to 87%. Every capture counts.
- **A free king is worth pawns.** Each corner your king can reach in two moves is worth six tenths of a pawn. Keep lines open around your king and close those of the enemy king.
- **The edge is a road.** On an edge, the king can only be captured along the edge: it is the safest place for it and the most dangerous for your opponent. Just beware of the square next to a corner (pattern 5.5).
- **Guarding the corners is not enough.** Pawns standing near the corners are worth almost nothing in themselves. What matters is being able to bring a piece onto the enemy king's line at the right moment.
- **Having the move is worth a fifth of a pawn.** Black, who plays first, scores 51.6% over 1370 games: a slight edge, within the margin of error.

3 The opening

The central cross. At the start, the pair is surrounded: black pawns on e6 and g6, white pawns on f5 and f7. These pawns hold each other. Moving one of them without calculation can lose a pawn at once (figure 2).

A flat opening. I evaluated each of Black's 96 first moves with a 7- then 8-move-deep search. Every move has a twin by a half-turn of the board, worth the same. 21 of the 48 distinct moves stay within half a pawn of the best: the first move decides nothing, but some moves are mistakes.



2 A typical first-move mistake by White (Black's first move, away from the centre, is not shown): e7-b7? frees e7, and Black plays e6-e7, capturing f7 between e7 and g7. White's three worst replies to Black's best first moves all move the e7 pawn.

Black's first move (and its twin)	Value for Black, in pawns
The best	
d1-d3 · h11-h9	+0.23
d1-c1 · h11-i11	+0.23
f10-c10 · f2-i2	+0.20
d11-c11 · h1-i1	+0.15
f10-i10 · f2-c2	+0.12
d11-d9 · h1-h3	+0.05
b10-b9 · j2-j3	+0.03
The worst	
d11-b11 · h1-j1	**−1.12**
e5-e4 · g7-g8	**−1.20**
b10-b7 · j2-j5	**−1.20**
e5-b5 · g7-j7	**−1.29**
b10-a10 · j2-k2	**−1.35**

- Do not move a pawn of the central cross (e6, g6 for Black; f5, f7 for White), or a pawn supporting it, without checking what it protects.
- Do not move a pawn next to an empty corner: a single enemy pawn can capture it (pattern 5.2). Two of the five worst first moves do exactly that; two others move e5, which supports the cross.
- Prefer moves of outer pawns, which prepare lines without weakening anything.
- The pair usually leaves f6 between move 18 and move 36 (median: move 24, both sides counted). Until then the game is a pawn battle.

4 Moving the pair out

This is the main strategic decision of the game: who takes his king out first, and when? The player who leaves takes the initiative but exposes his king. The player who stays on f6 cannot be captured as long as his king does not move.

Over all games, the player who leaves first scores 52%: neither role is favoured. It all depends on the situation at the moment you leave.

When the pair is moved out	Frequency	Score of the player who leaves
Fewer pawns than the opponent	40%	40%
Same number of pawns	26%	50%
More pawns	33%	68%

Who wins, and how	Share of wins
The player who left first, by escaping to a corner	45%
The player who left first, by capture	7%
The player who stayed on f6, by escaping to a corner	31%
The player who stayed on f6, by capture	17%

> **THE KEY IDEA**
>
> Move the pair out when you are ahead in material: you then convert two games out of three, usually with a race to a corner. When behind, do not leave on a gamble: stay on f6, where your king cannot be captured, and try to catch the enemy king when it ventures into the middle.

5 Tactical patterns

5.1 · The pincer

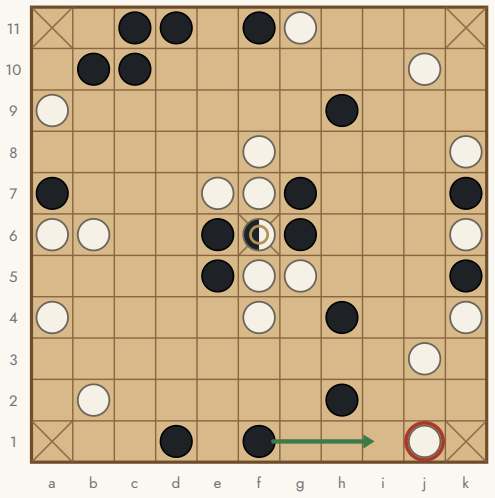
The basic weapon: 94% of captures are made between the arriving pawn and a friendly pawn, about 13 per game. A piece is in danger as soon as an enemy pawn can reach the empty square on its other side.

5.2 · The empty anvil

An empty corner or the empty throne counts as a friendly pawn: 3.4% of captures are made against a corner and 2.2% against f6. A pawn standing next to an empty corner falls to a single enemy pawn (figure 3).

5.3 · The double capture

A pawn arriving between two well-supported enemy pieces captures both. Rare, but 18% of games contain at least one.

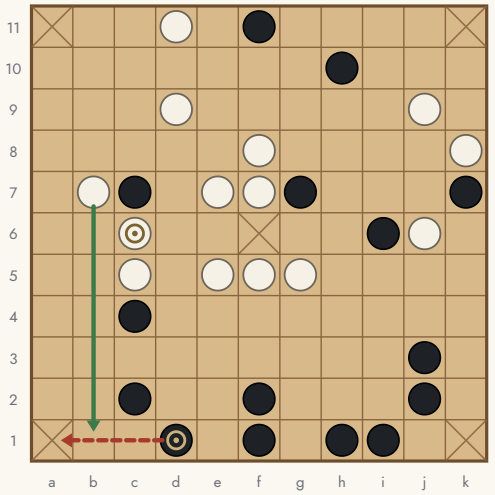


3 From a real game. f1-i1 captures j1: the empty corner k1 is the other side.

5.4 · Threat and defence

A winning threat cuts your opponent's options from 114 possible moves to 4.7 on average, and to a single move in 23% of cases (figure 4). It is the tool for gaining time. The defences: the threatened king steps aside (60%), a piece closes the line (19%), a piece occupies the capture square (16%), or a capture removes the threat (5%).

When attacking, a threat that leaves a single move lets you predict the reply, and so calculate far ahead. Chain forcing threats: each defence you impose is a tempo gained to bring one more piece. **When defending,** look for the defence that also improves your position: a capture, or a piece coming from an area where it was useless.

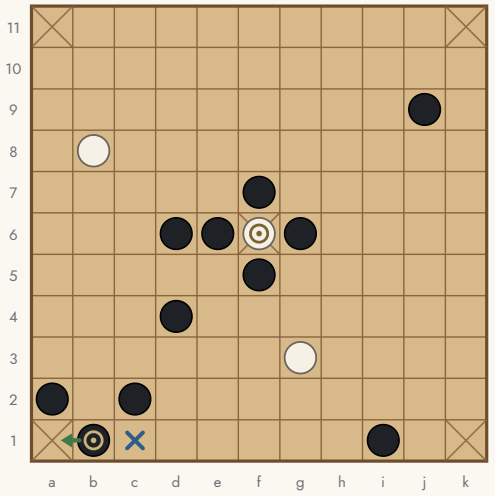


4 From a real game. The black king threatens a1. Only one of White's 95 moves is allowed: b7-b1.

5.5 · The corner as an anvil against the king

A king standing right next to a corner can be captured by a single enemy pawn: the empty corner counts as the other side. Before bringing your king there, check that no enemy pawn can reach the square on its other side (figure 5).

The pattern works both ways. An enemy pawn able to reach the second edge square from a corner (c1 for corner a1) forbids your king from stopping next to that corner (b1): the threat rule makes that move illegal. Conversely, one of your pawns placed that way is enough to hold a corner against the enemy king.

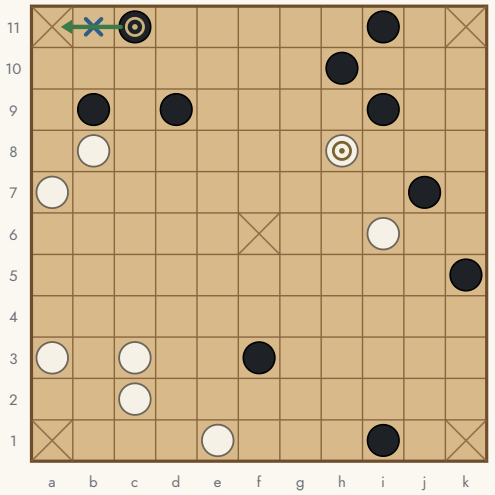


5 End of a real game. The black king on b1 wins with b1-a1. A white pawn on c1 would have captured it against the corner, but none can get there.

5.6 · The unstoppable line

96% of escape wins come from a single threat that nothing can stop: no enemy piece can reach the line between the king and the corner, and the king cannot be captured (figure 6).

The method: for each line between the king and a corner, list the empty squares and, for each one, the enemy pieces able to reach it in one move. If there are none and the king cannot be captured, the line is unstoppable. Do this count for your king, and above all for your opponent's.

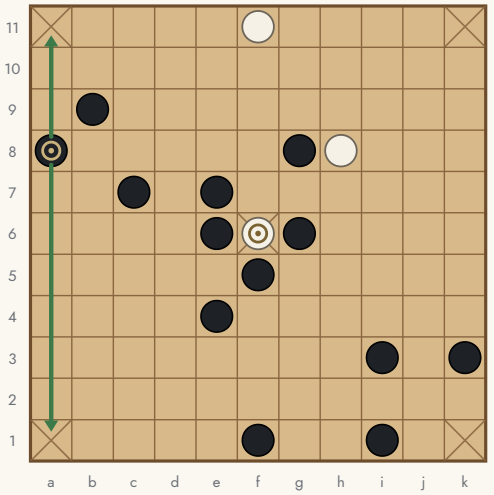


6 End of a real game. The black king threatens a11 through b11, a square no white piece can reach.

5.7 · The corner fork

A king on an edge that is open in both directions threatens two corners at once. Only one line can be closed: the game is over. This is how 4.3% of escape wins happen (figure 7).

A fork is born on an open edge. Before bringing your king onto an edge, count the enemy pieces on that edge; when defending, keep at least one piece on every edge the enemy king can reach.

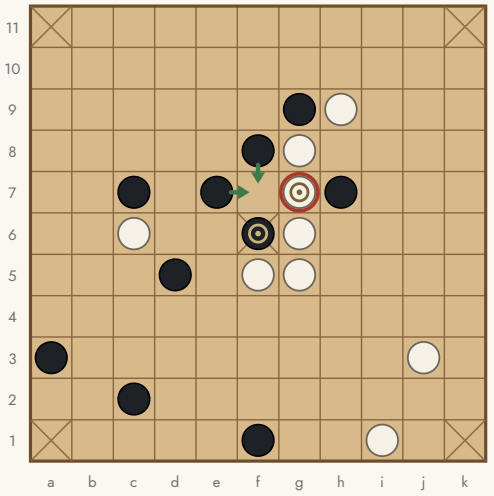


7 End of a real game. The black king on a8 threatens a11 and a1: White can close only one line.

5.8 · The pincer on the king

82% of capture wins come from at least two moves that capture the king, often two pawns aiming at the same square. The other side is a friendly pawn in 94% of cases, the empty throne in the rest (figure 8).

Away from f6, the king can be captured in two directions. Put a pawn on a square next to the king, then bring two pawns aiming at the opposite square: your opponent must move his king or occupy that square. If he can do neither, the game is won.



8 End of a real game. The white king on g7 has h7 behind it. Both e7-f7 and f8-f7 capture it, and White cannot occupy f7.

6 Viable plans

Plan A · The race to a corner

The most frequent winning plan (77% of wins). Move the pair out, keep your king inside, away from pincers, and prepare a line to an edge that your opponent will no longer be able to close. The winning king makes a median of 4 moves, and the final strike is sudden. This plan works even when down in material: 32% of escape winners had fewer pawns at the end. In 59% of cases, the winner is the player who moved the pair out.

Plan B · Hunting the king

22% of wins. Keep your king on f6, where it cannot be captured, win material (71% of capture winners had more pawns at the end), then catch the enemy king as it crosses the middle of the board. In 69% of cases, the hunter is the player who stayed on f6. The key: prepare two pawns aiming at the same square next to the enemy king, with a support on the other side.

Plan C · Suffocation

Immobilising every enemy piece never happened in the games analysed. With the threat rule, your opponent almost always keeps a free pawn move. It is not a viable plan.

Choosing your plan. If you are ahead in pawns, leave and race to an edge: the race converts well. If you are behind, stay on f6 and hunt: the king that leaves first often ends up crossing the middle, where it gets caught.

7 Before every move

1. **Lines to the corners.** Can the enemy king reach an edge, then a corner, in one or two moves? Which pieces could close those lines?
2. **My king.** Can it be captured next move? Look for the neighbouring squares two enemy pawns can reach.
3. **My pawns under attack.** A pawn next to an empty corner, or between an enemy piece and a reachable empty square, is in danger.
4. **The threats I can make.** A threat forces your opponent to defend: he is left with only a few allowed moves. Chain them to gain time.
5. **The pair.** Do you have more pawns? Then leaving may be the right moment. Otherwise keep your king on f6.
6. **Calculate.** In 24% of positions, a move chosen by looking only two moves ahead loses at least a pawn compared with the best one. The game rewards calculation.

> **IN SHORT**
>
> 1. Every pawn counts: one pawn ahead adds about nine percentage points to your winning chances.
> 2. Games are almost always won by escaping to a corner: watch the lines, not the corners.
> 3. The winning king stays inside and jumps onto the edge only at the last moment.
> 4. On f6 your king cannot be captured: leave only with an advantage, or when it is safe.
> 5. Next to a corner, a single pawn is enough to capture: your pawns and your king alike.
> 6. A threat leaves your opponent few moves: use it to gain time.
> 7. Two threats at once win the game.

8 Method and limits

The games were played by an alpha-beta AI whose rules were checked against the Ludii game engine. Openings were slightly randomised, then the AI played its best moves with a budget of 60,000 to 1,000,000 positions per move. The asset values come from a logistic regression on 38,307 positions. These figures describe computer play: at a high human level the proportions may differ, but the patterns stay the same.